Appendices E and F of `main.tex` compared with JIMWLK⊗LO and BFKL⊗LO

Result.

- **Appendix E (BFKL), Eq. (LOBFKL): correct as written.** In the colour-singlet projection $\langle\rho^a(x)\rho^b(y)\rangle = \delta^{ab}\langle\rho^f(x)\rho^f(y)\rangle/(N_c^2-1)$ it equals BFKL⊗LO of `Evolution_from_final.pdf`, Eq. (6.2), exactly. This holds for symmetric and for general adjoint Wilson lines.
- **Appendix F (JIMWLK): two corrections needed**. They are the same two as reported before, and are still in the pasted version:
  1. In $dN^{(1)}$–$dN^{(4)}$ the matrix $(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))$ (or its transpose) is written in every line. It must be $(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))$ at the points u,v where the two J_R act, i.e. the points in $\mathcal{K}(u,v,t)$ of that line. In Eq. (HLOfinal) this matrix sits next to $J_R^c(u)J_R^d(v)$. After the commutators, the δ-functions set u and v to x,w,y,w'; the matrix must follow them.
  2. The overall sign of $dN^{(5)}$ is $+$. Eq. (HLOfinal) is right: $[(U^\dagger(u)-U^\dagger(t))U(u)T^c]_{cd} = -\mathrm{tr}[(U^\dagger(u)-U^\dagger(t))U(u)T^d]$. Together with $d/dY=-H$ the two minus signs give $+$.

  With both corrections, Appendix F equals JIMWLK⊗LO of `Evolution_from_final.pdf`, Eq. (3.2), exactly. With only one of them it does not.

The corrected Appendix F follows; changed factors are blue. The prefactor is

$$P \equiv \frac{\alpha_s^2\,\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\,\langle\rho^f(x)\rho^f(y)\rangle.$$

$$\begin{aligned}\frac{dN^{(1)}}{d^2w\,d^2w'\,dy} = P\Big[&\mathcal{K}(y,x,t)\,\mathrm{Tr}\big\{T^eU^\dagger(x)U(y)T^e\,(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\big\}\\ &-\mathcal{K}(w',x,t)\,\mathrm{Tr}\big\{T^eU^\dagger(x)U(w')T^e\,(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\big\}\\ &-\mathcal{K}(y,w,t)\,\mathrm{Tr}\big\{T^eU^\dagger(w)U(y)T^e\,(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\big\}\\ &+\mathcal{K}(w',w,t)\,\mathrm{Tr}\big\{T^eU^\dagger(w)U(w')T^e\,(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\big\}\Big]\end{aligned}\tag{1}$$

$$\begin{aligned}\frac{dN^{(2)}}{d^2w\,d^2w'\,dy} = P\Big[&\mathcal{K}(x,y,t)\,\mathrm{Tr}\big\{T^eU^\dagger(x)U(y)T^e\,(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\big\}\\ &-\mathcal{K}(x,w',t)\,\mathrm{Tr}\big\{T^eU^\dagger(x)U(w')T^e\,(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\big\}\\ &-\mathcal{K}(w,y,t)\,\mathrm{Tr}\big\{T^eU^\dagger(w)U(y)T^e\,(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\big\}\\ &+\mathcal{K}(w,w',t)\,\mathrm{Tr}\big\{T^eU^\dagger(w)U(w')T^e\,(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\big\}\Big]\end{aligned}\tag{2}$$

$$\begin{aligned}\frac{dN^{(3)}}{d^2w\,d^2w'\,dy} = -P\Big[&\mathcal{K}(y,y,t)\,\mathrm{Tr}\big\{T^e(U^\dagger(x)-U^\dagger(w))U(y)T^e\,(U^\dagger(y)-U^\dagger(t))(U(y)-U(t))\big\}\\ &-\mathcal{K}(w',w',t)\,\mathrm{Tr}\big\{T^e(U^\dagger(x)-U^\dagger(w))U(w')T^e\,(U^\dagger(w')-U^\dagger(t))(U(w')-U(t))\big\}\Big]\end{aligned}\tag{3}$$

$$\begin{aligned}\frac{dN^{(4)}}{d^2w\,d^2w'\,dy} = -P\Big[&\mathcal{K}(x,x,t)\,\mathrm{Tr}\big\{T^eU^\dagger(x)(U(y)-U(w'))T^e\,(U^\dagger(x)-U^\dagger(t))(U(x)-U(t))\big\}\\ &-\mathcal{K}(w,w,t)\,\mathrm{Tr}\big\{T^eU^\dagger(w)(U(y)-U(w'))T^e\,(U^\dagger(w)-U^\dagger(t))(U(w)-U(t))\big\}\Big]\end{aligned}\tag{4}$$

$$\begin{aligned}\frac{dN^{(5)}}{d^2w\,d^2w'\,dy} = +P\Big[&\mathcal{K}(y,y,t)\,\mathrm{Tr}\big\{(U^\dagger(y)-U^\dagger(t))U(y)T^d\big\}\,\mathrm{Tr}\big\{(U^\dagger(x)-U^\dagger(w))U(y)T^d\big\}\\ &-\mathcal{K}(w',w',t)\,\mathrm{Tr}\big\{(U^\dagger(w')-U^\dagger(t))U(w')T^d\big\}\,\mathrm{Tr}\big\{(U^\dagger(x)-U^\dagger(w))U(w')T^d\big\}\\ &-\mathcal{K}(x,x,t)\,\mathrm{Tr}\big\{(U^\dagger(x)-U^\dagger(t))U(x)T^d\big\}\,\mathrm{Tr}\big\{U^\dagger(x)(U(y)-U(w'))T^d\big\}\\ &+\mathcal{K}(w,w,t)\,\mathrm{Tr}\big\{(U^\dagger(w)-U^\dagger(t))U(w)T^d\big\}\,\mathrm{Tr}\big\{U^\dagger(w)(U(y)-U(w'))T^d\big\}\Big]\end{aligned}\tag{5}$$

and $dN/d^2w\,d^2w'\,dy = \sum_{n=1}^5 dN^{(n)}$.

- In $dN^{(1)}$ the matrix is $(U^\dagger(u)-U^\dagger(t))(U(v)-U(t))$ for $\mathcal{K}(u,v,t)$.
- In $dN^{(2)}$ it is the transpose, $(U^\dagger(v)-U^\dagger(t))(U(u)-U(t))$ for $\mathcal{K}(u,v,t)$, because there $\mathrm{Tr}[WT^dT^c]M_{cd} = \mathrm{Tr}[T^eWT^eM^T]$.
- In $dN^{(3)}$, $dN^{(4)}$ it is $(U^\dagger(v)-U^\dagger(t))(U(v)-U(t))$ for $\mathcal{K}(v,v,t)$.
- The first line of $dN^{(2)}$ was already right as printed.

Numbers. Each value is the integrand of $dN/d^2w\,d^2w'\,dy$ at a random point (x, y, w, w', t), including $\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')$, in units of $\alpha_s^2\delta Y\langle\rho^f(x)\rho^f(y)\rangle/(8\pi^6(N_c^2-1))$, symmetrized over $x\leftrightarrow y$. "Note" is the written-out result of `Evolution_from_final.pdf`, projected on the colour singlet. Appendix E is $-N_c\times$ the bracket of Eq. (LOBFKL).

U	seed	JIMWLK (note)	App. F as pasted	App. F corrected	BFKL (note)	App. E
symmetric	1	−47.715089	−28.081101	−47.715089	79.307329	79.307329
symmetric	2	−230.981876	−179.490327	−230.981876	−107.565523	−107.565523
symmetric	3	0.016879	0.012758	0.016879	0.089954	0.089954
symmetric	4	31.433670	15.846990	31.433670	−12.054752	−12.054752
general	1	4.223482	5.866983	4.223482	95.247533	95.247533
general	2	88.007301	52.780006	88.007301	91.646271	91.646271
general	3	0.014910	0.007840	0.014910	0.133215	0.133215
general	4	17.797915	7.705099	17.797915	−25.710656	−25.710656